

Experimental Petrology: A Laboratory-based Method for Studying Processes in Planetary Interiors and Understanding their Expressions in the Rock Record

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University of Tennessee**



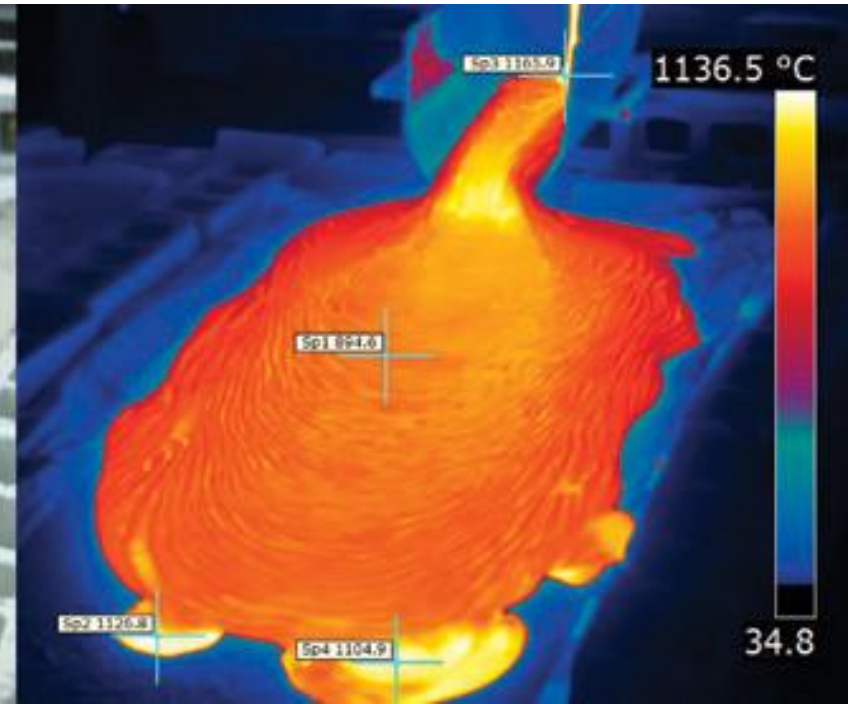
Petrology is the study of the compositions of rocks. The subject of this talk is experimental petrology, which generally applies to investigations of igneous rocks

Volcanic eruption, Fagradalsfjall (Iceland), 2021



Matthew Eisman, Getty

Lava pour experiment, Syracuse University



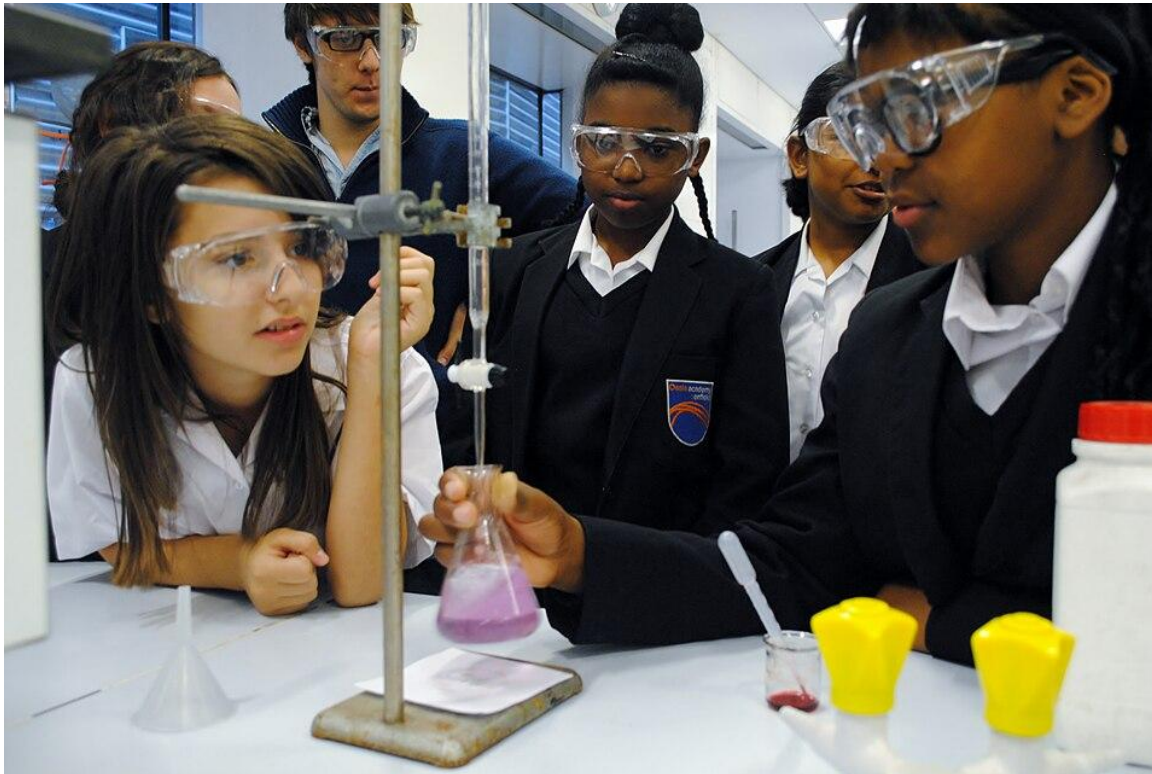
Earth magazine, Jeffrey Karson and Robert Wysocki

Outline

- What is experimental petrology?
- How can experimental petrology be used to answer science questions?
- What are the different apparatuses used in experimental petrology?
- Models are how individual experimental data are understood.
- What are the limitations of AI-based predictive model development?
- How can experimentally-parameterized models be used to address science questions?
- All models are wrong but some are useful.

Compositions of igneous rocks are measured by analytical methods (traditionally wet chemistry, and more recently, mass spectrometry)

A titration demonstration



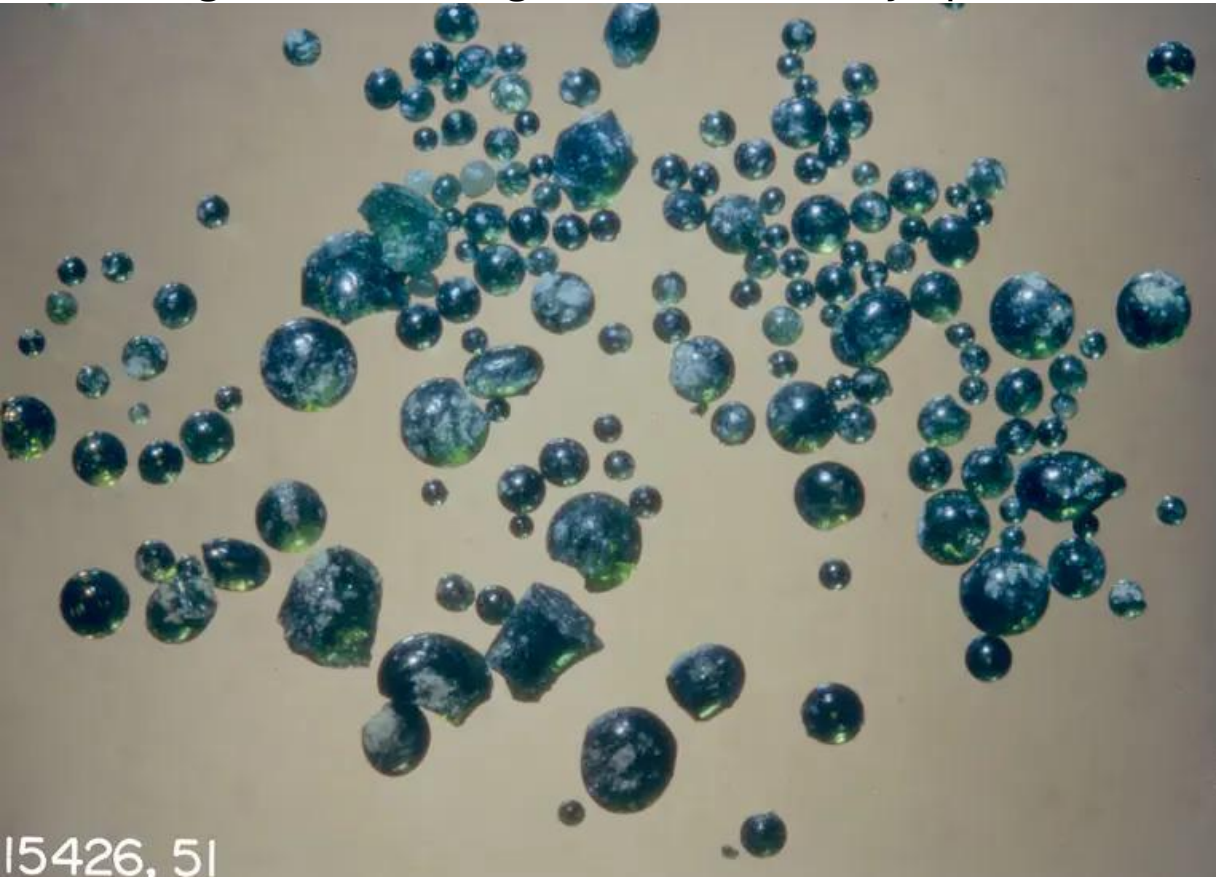
University College London

The JEOL electron microprobe at the University of Tennessee

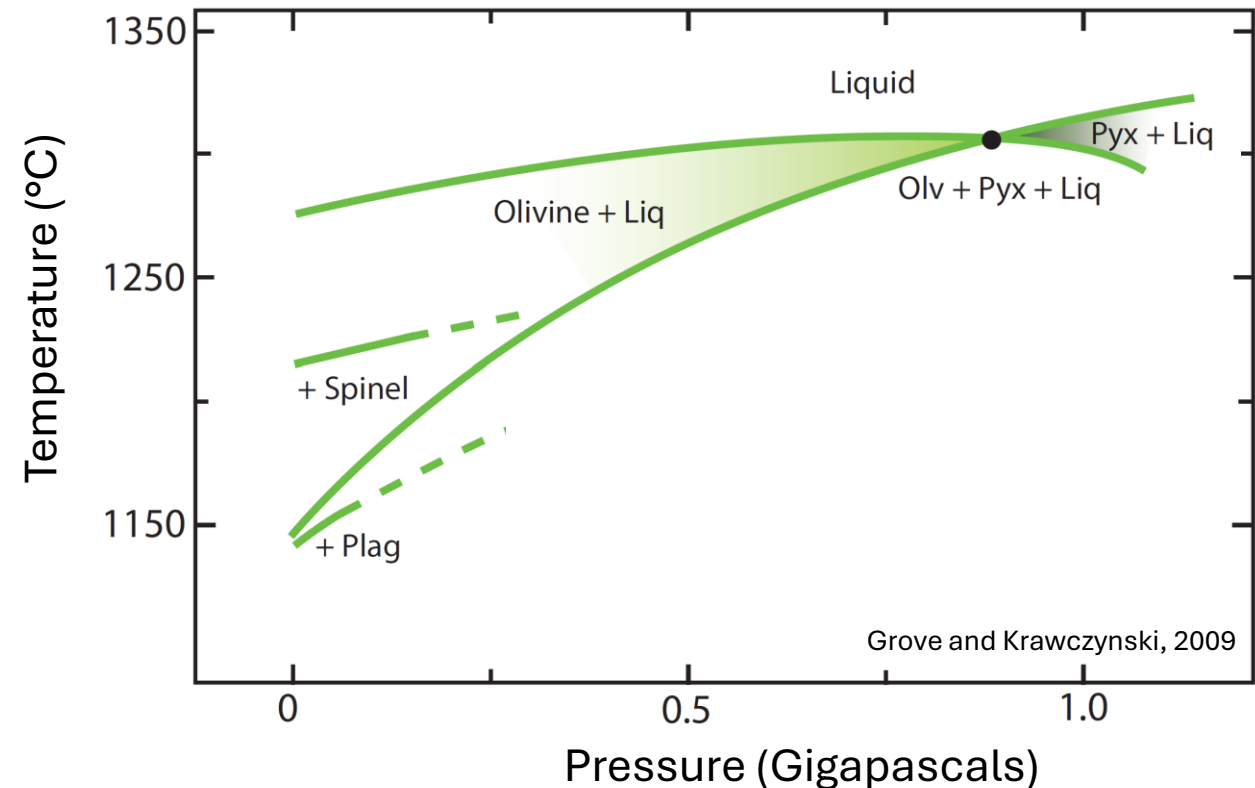


The compositions have no context without models of how the rocks form. The only way to develop such models is by experimentation

Lunar green volcanic glasses returned by Apollo 15

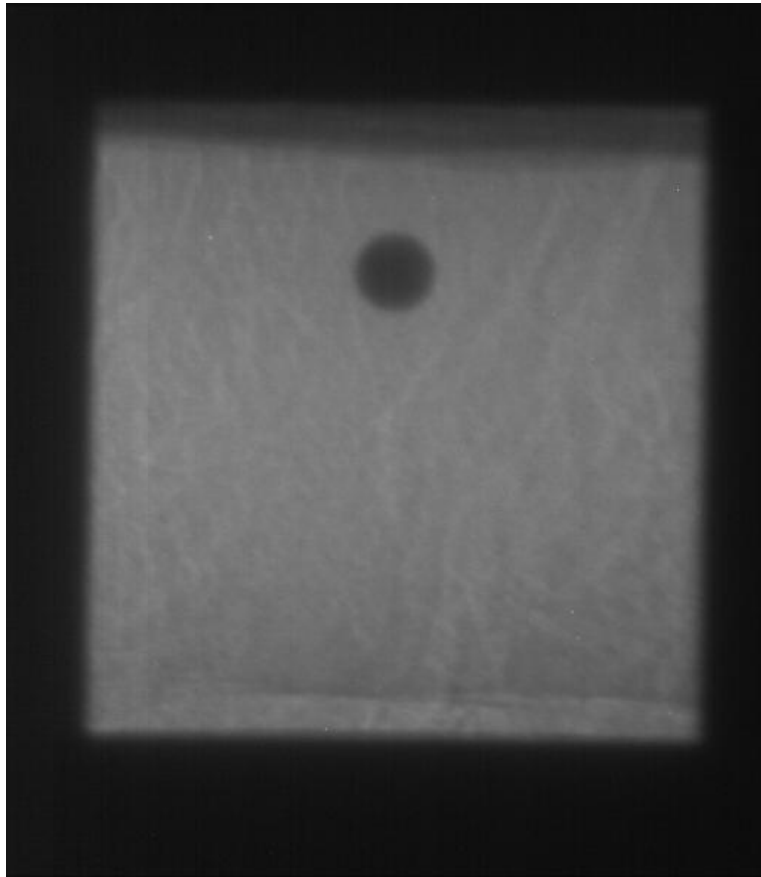


A summary of experiments on lunar green glasses which reveal the pressure and temperature of their origin

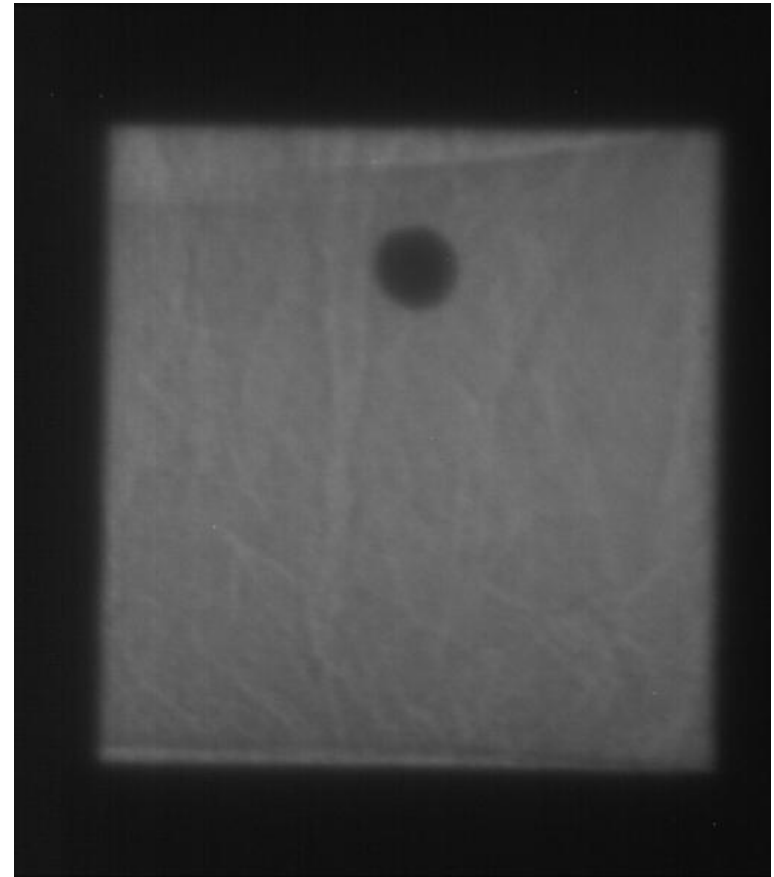


Experiments characterize physical properties of silicate liquids, such as viscosity

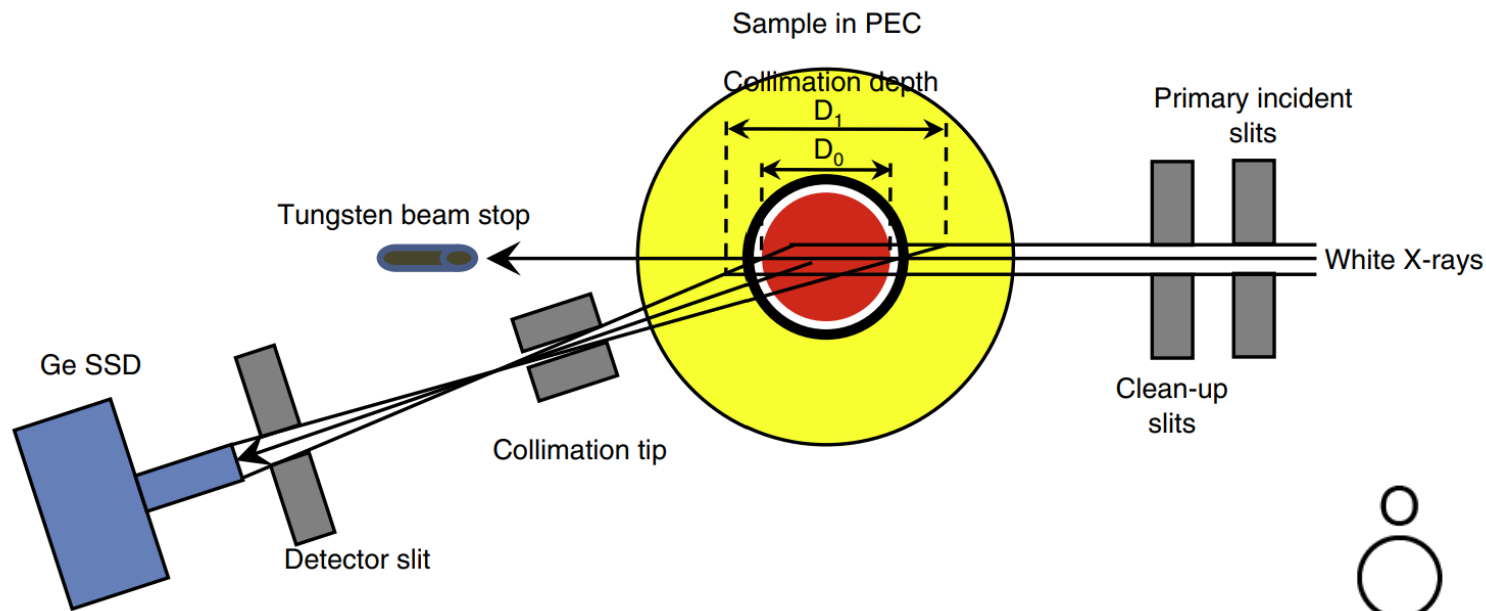
1300°C
 $\mu = 1.45 \pm 0.21 \text{ Pa}\cdot\text{s}$



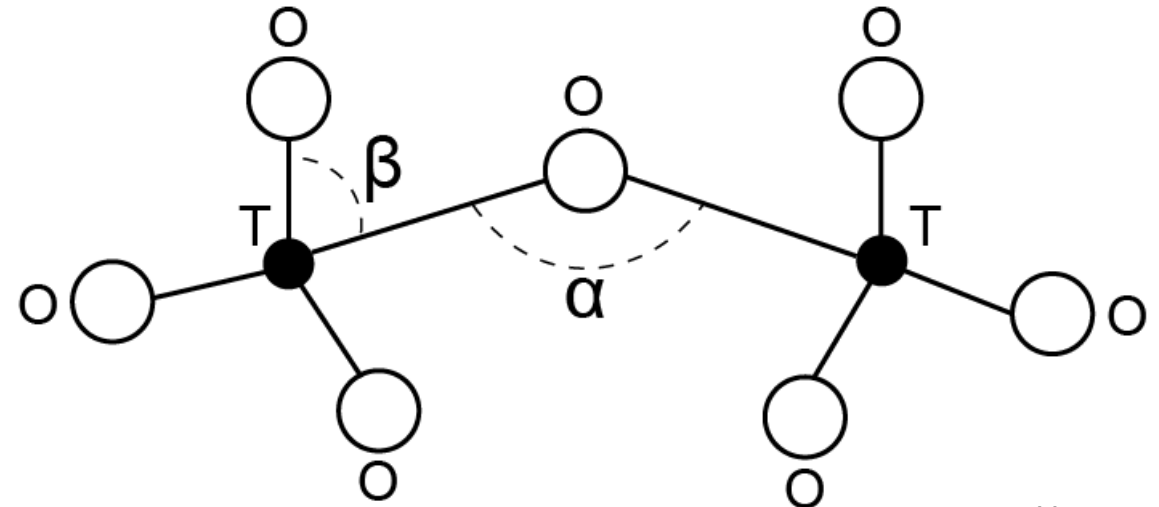
1600°C
 $\mu = 0.25 \pm 0.16 \text{ Pa}\cdot\text{s}$



Experiments characterize structural properties of silicate liquids

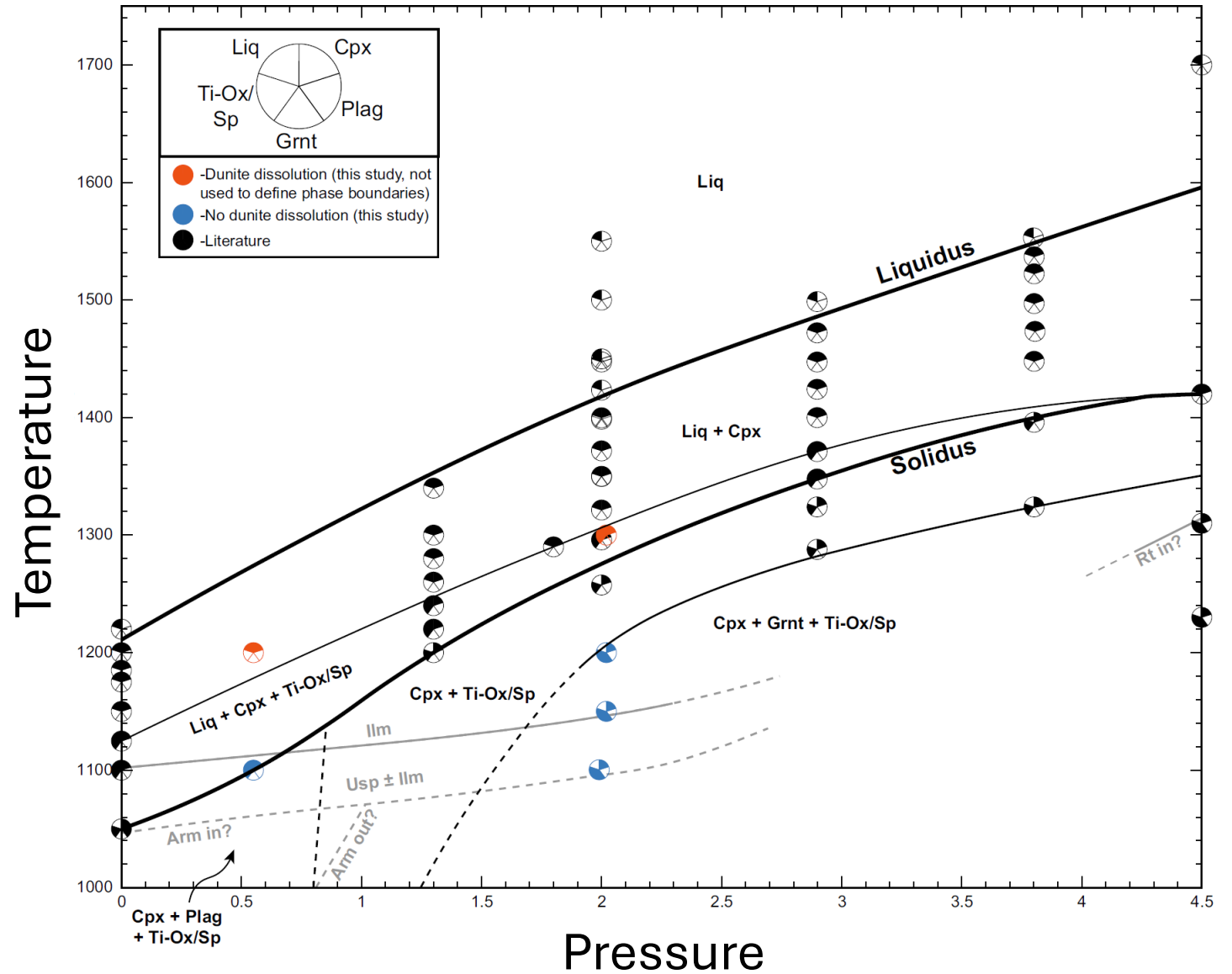


Kono et al., 2014



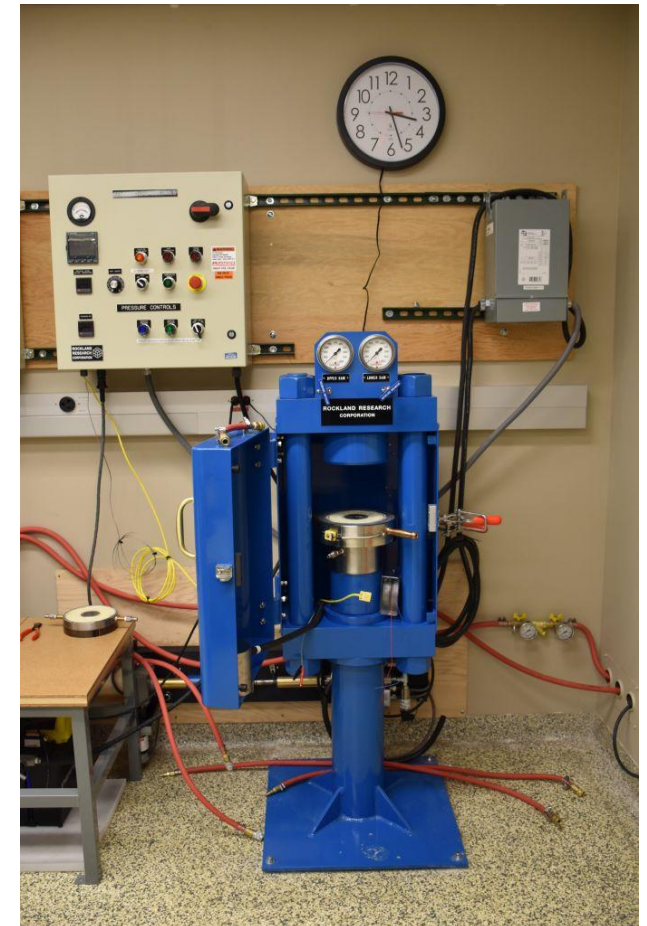
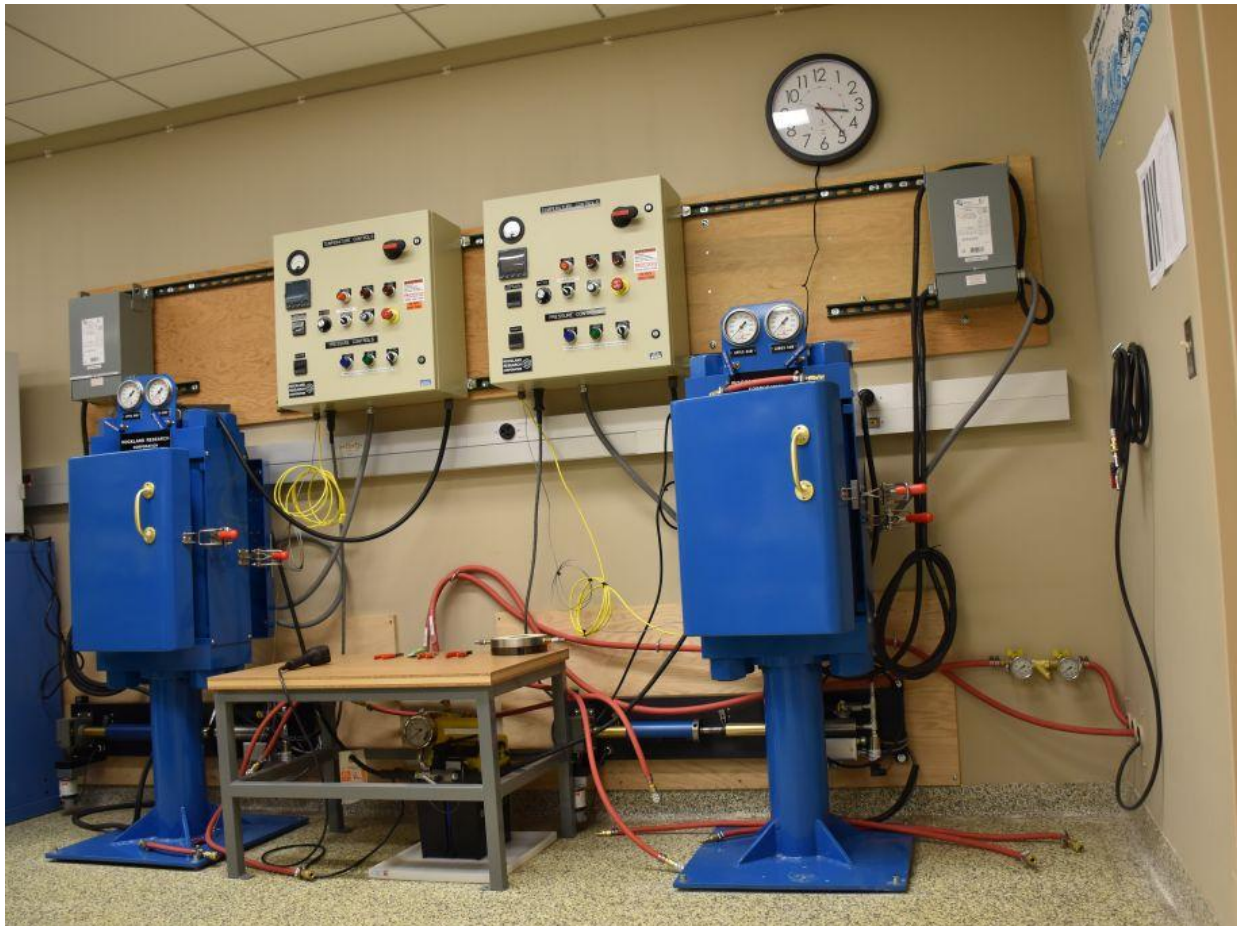
Mouser, 2022

Experiments characterize phase equilibria in silicate liquids (the sequence of minerals that precipitate from cooling magmas, and their compositions)



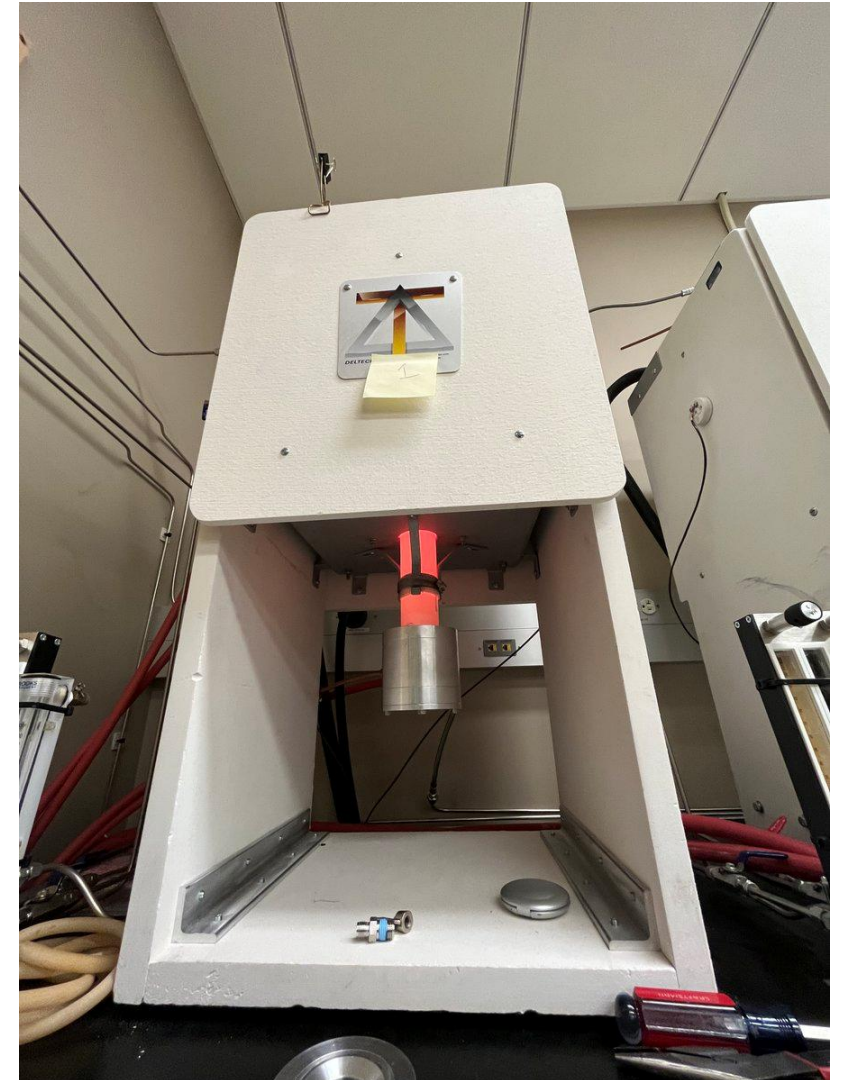
The workhorse apparatus in experimental petrology is the piston cylinder

Piston cylinder lab at the University of Tennessee



Another widely used apparatus is the gas mixing furnace

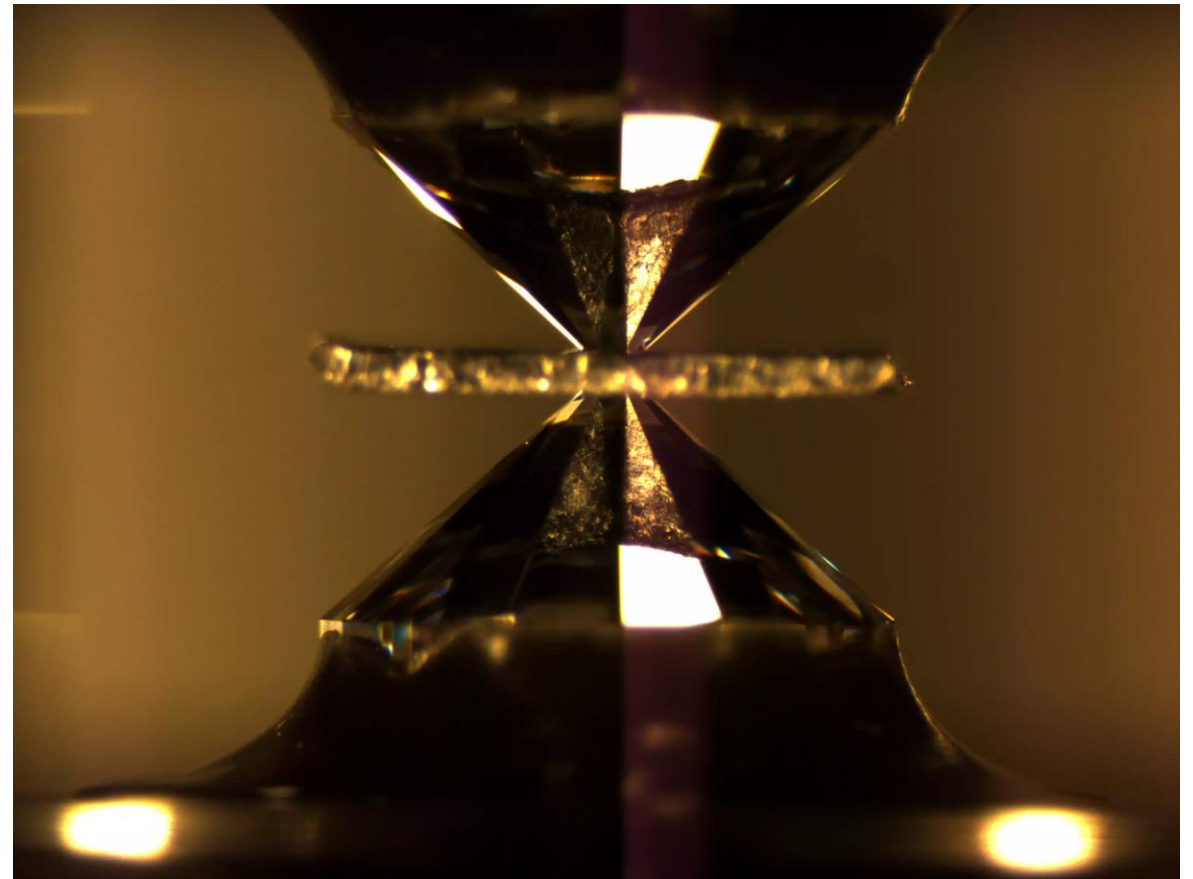
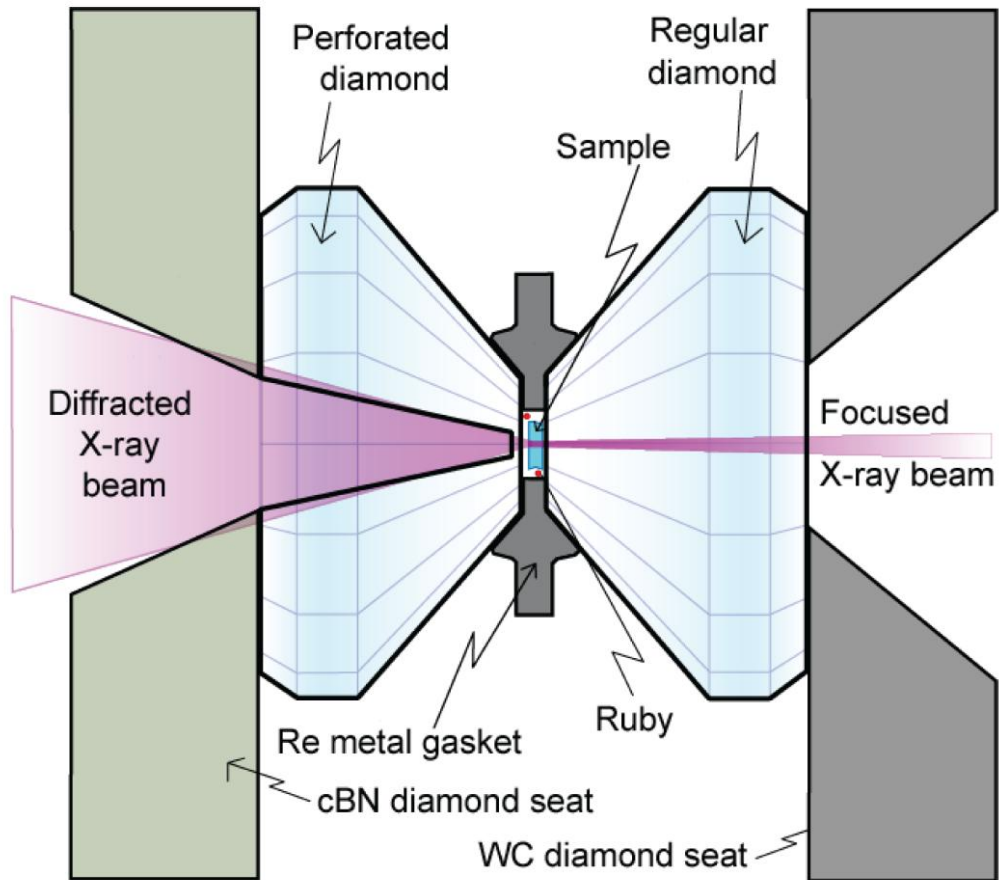
Gas mixing furnaces at the University of Tennessee



The multi anvil is used to investigate phase equilibria under higher pressures



Diamond anvils are used for even higher pressure-temperature experiments and can reach conditions relevant to Earth's core

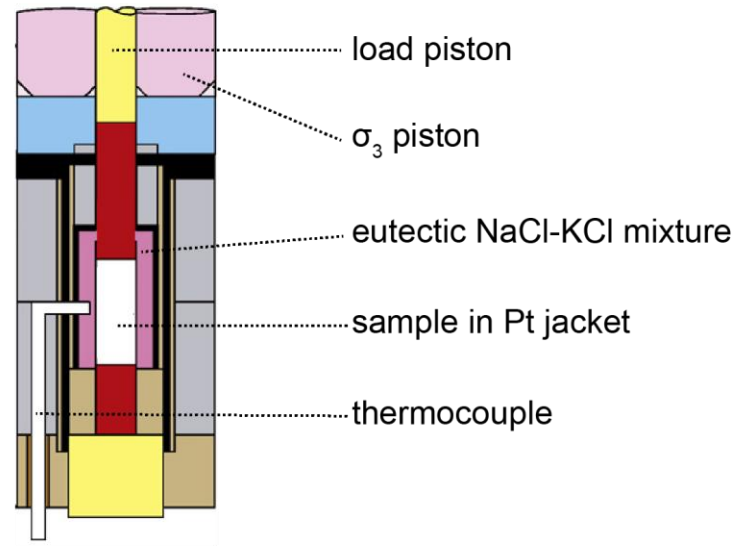
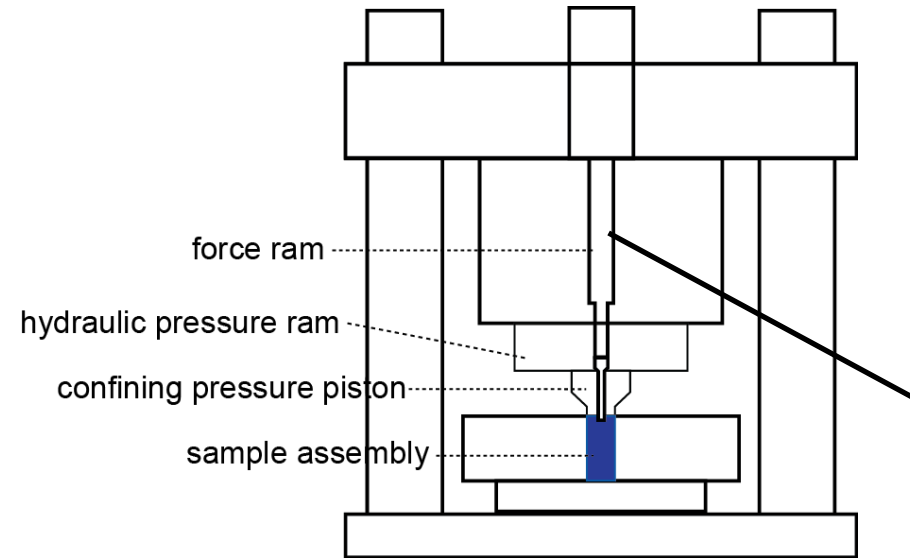


Another class of experimental apparatuses measure rheological properties of solid minerals and silicate liquids

Paterson apparatus, Hemholtz Centre for Geosciences, Potsdam

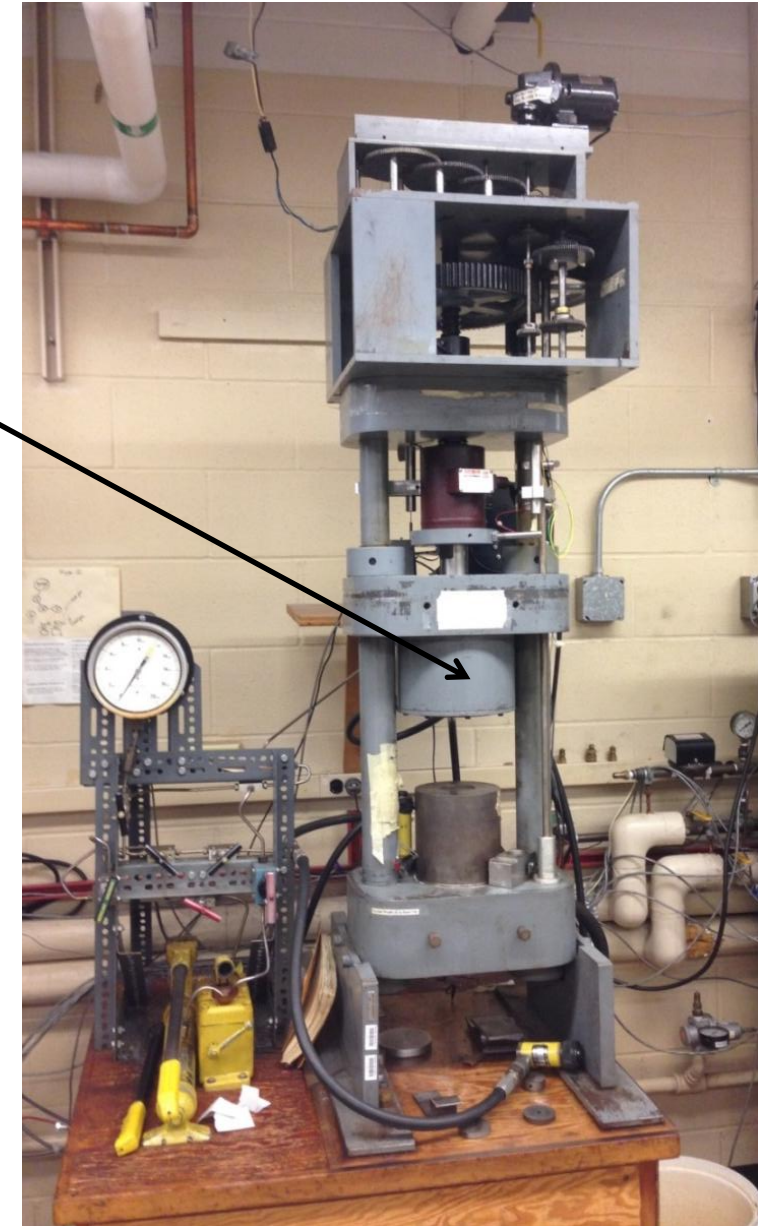


The Griggs apparatus is useful for studying deformation properties of minerals under planetary interior conditions

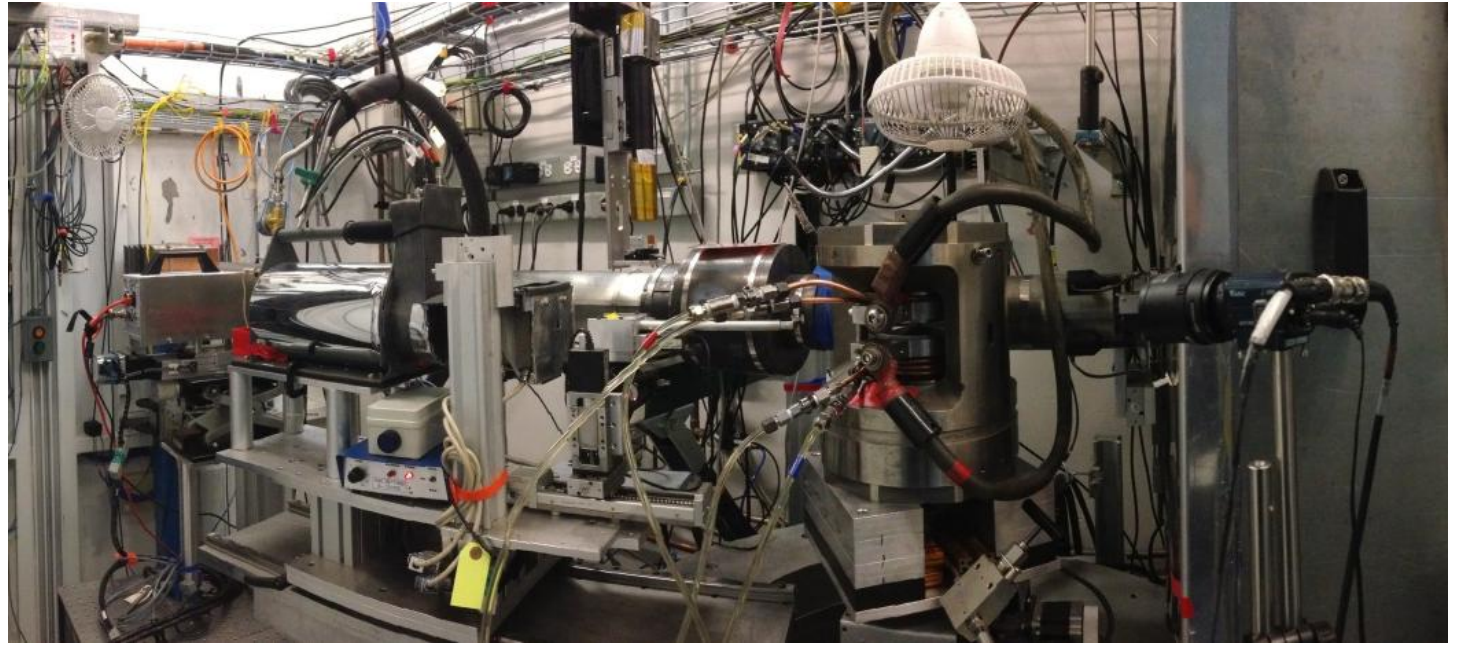


Tokle et al., 2021

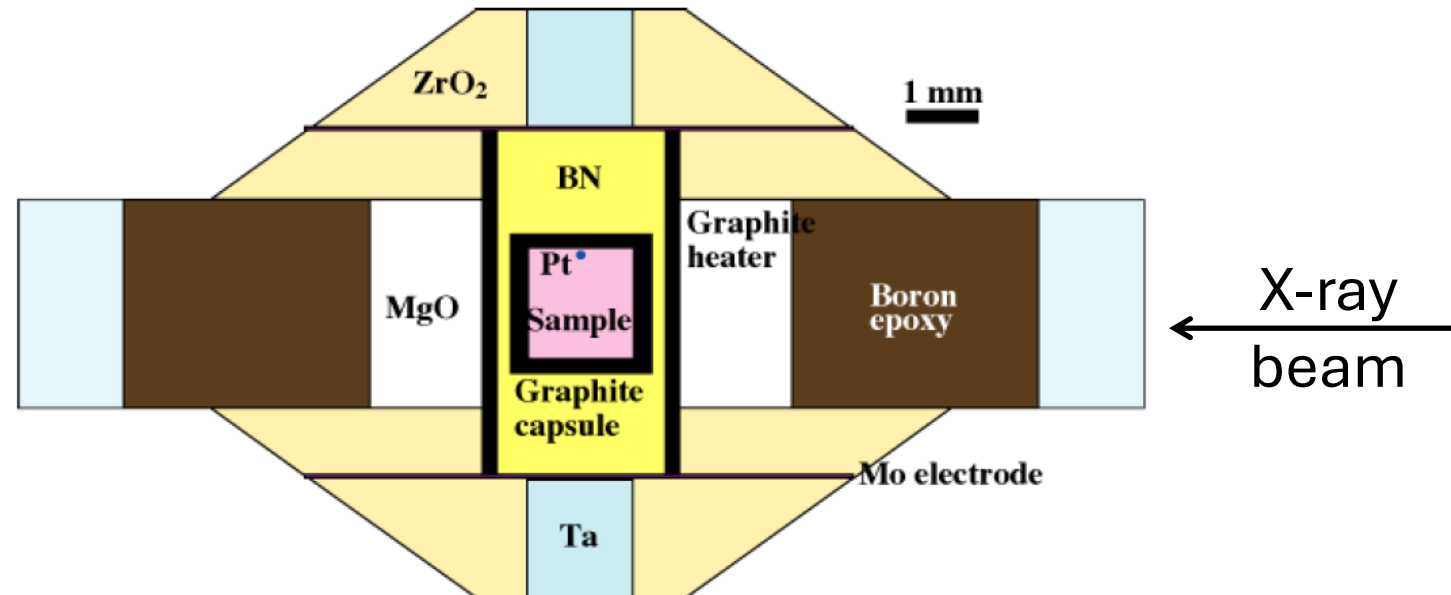
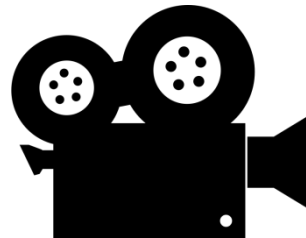
Endloaded Griggs apparatus at Brown University



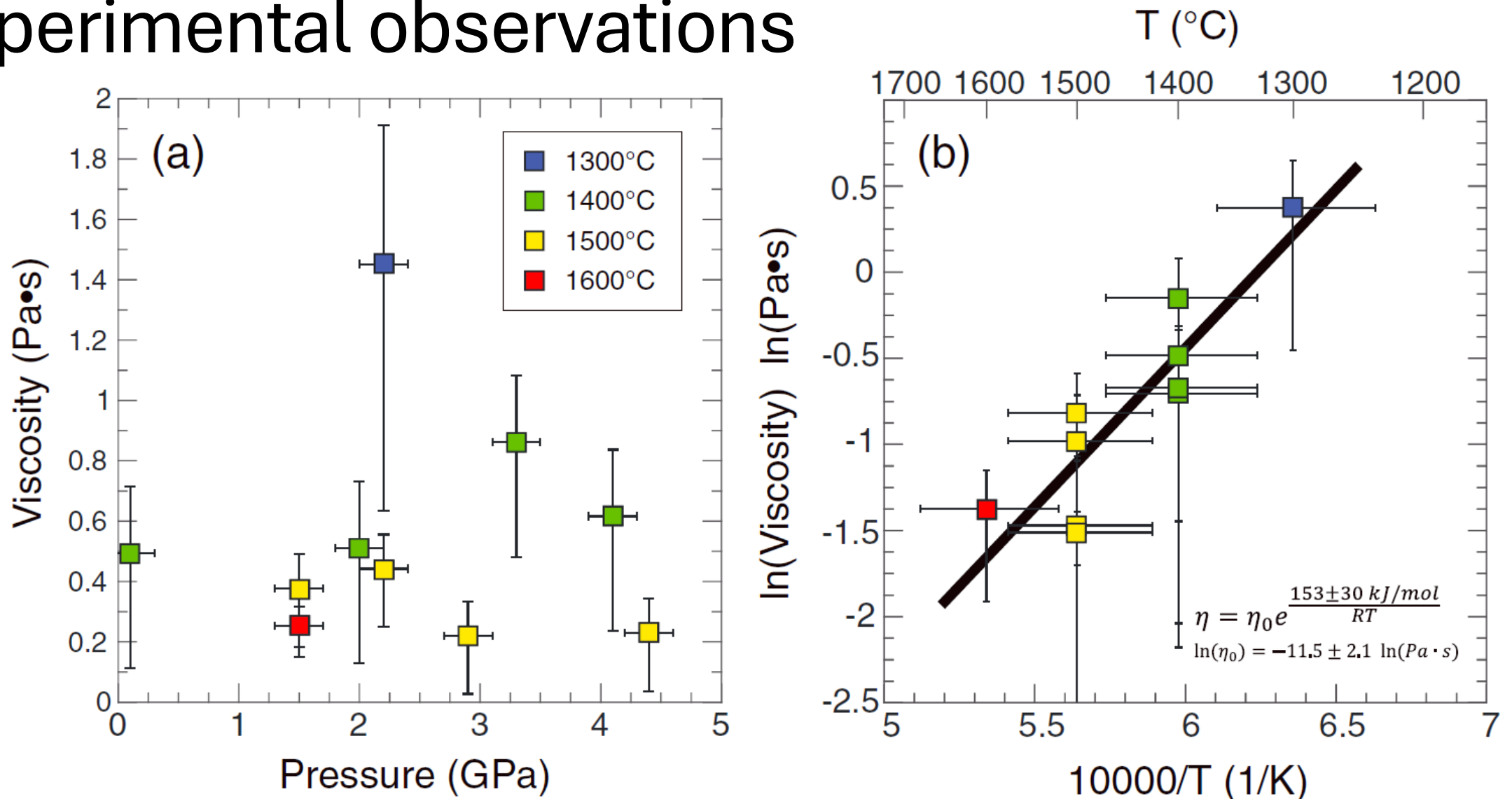
The Paris-Edinburgh apparatus is used to characterize the viscosity and structure of silicate liquids under high pressure-temperature conditions



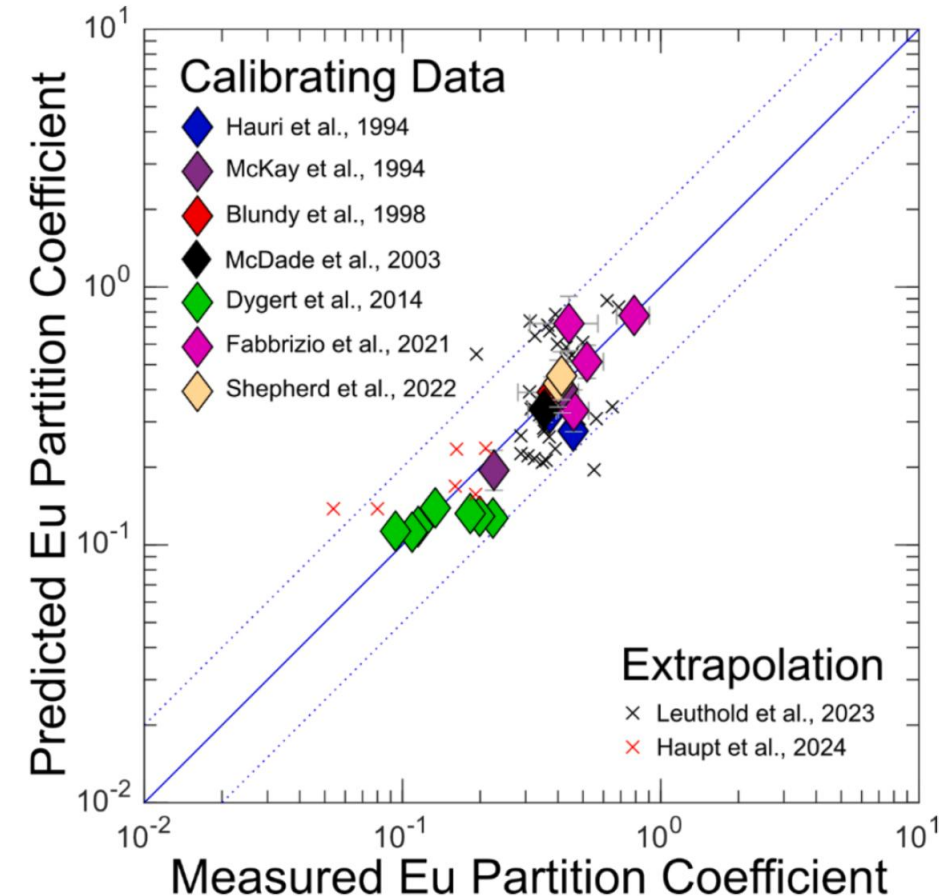
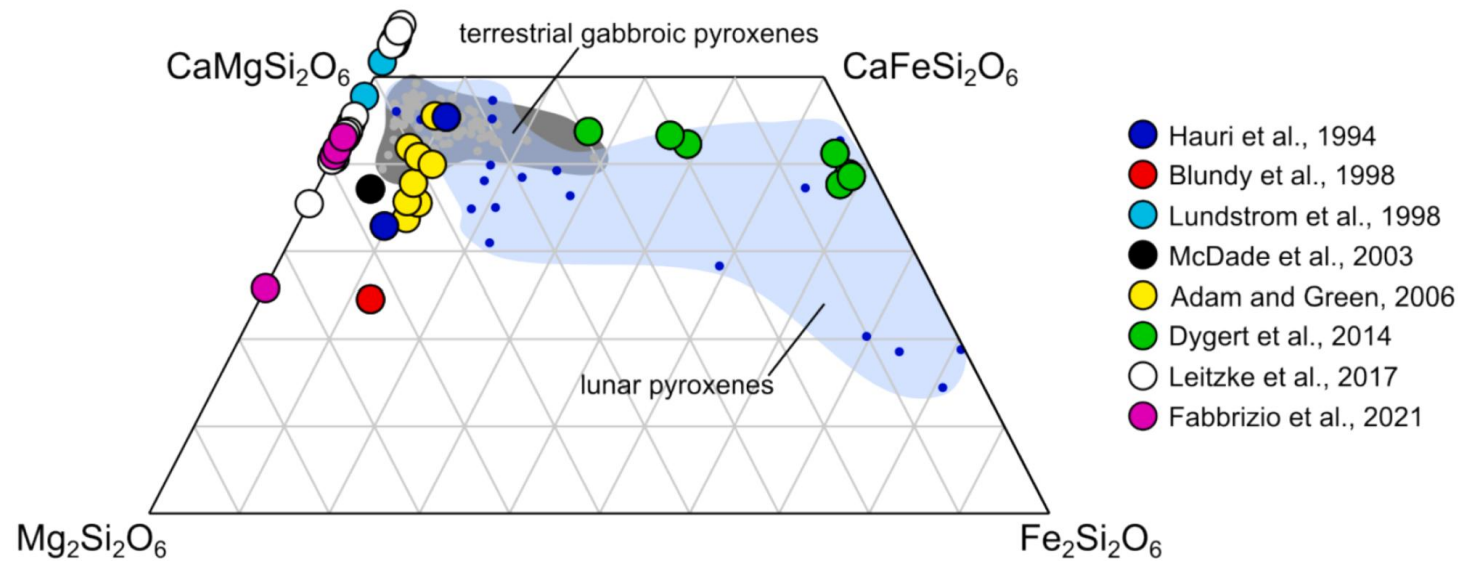
Paris-Edinburgh assembly



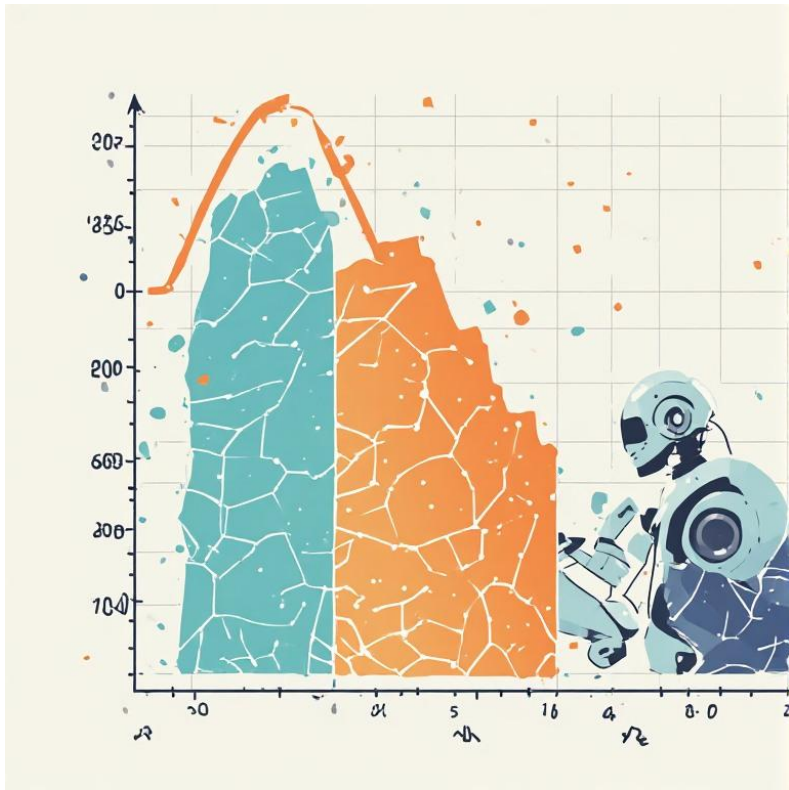
Simple models are developed to describe experimental observations



When a large number of experimental data accumulate, development of generalized models is a goal



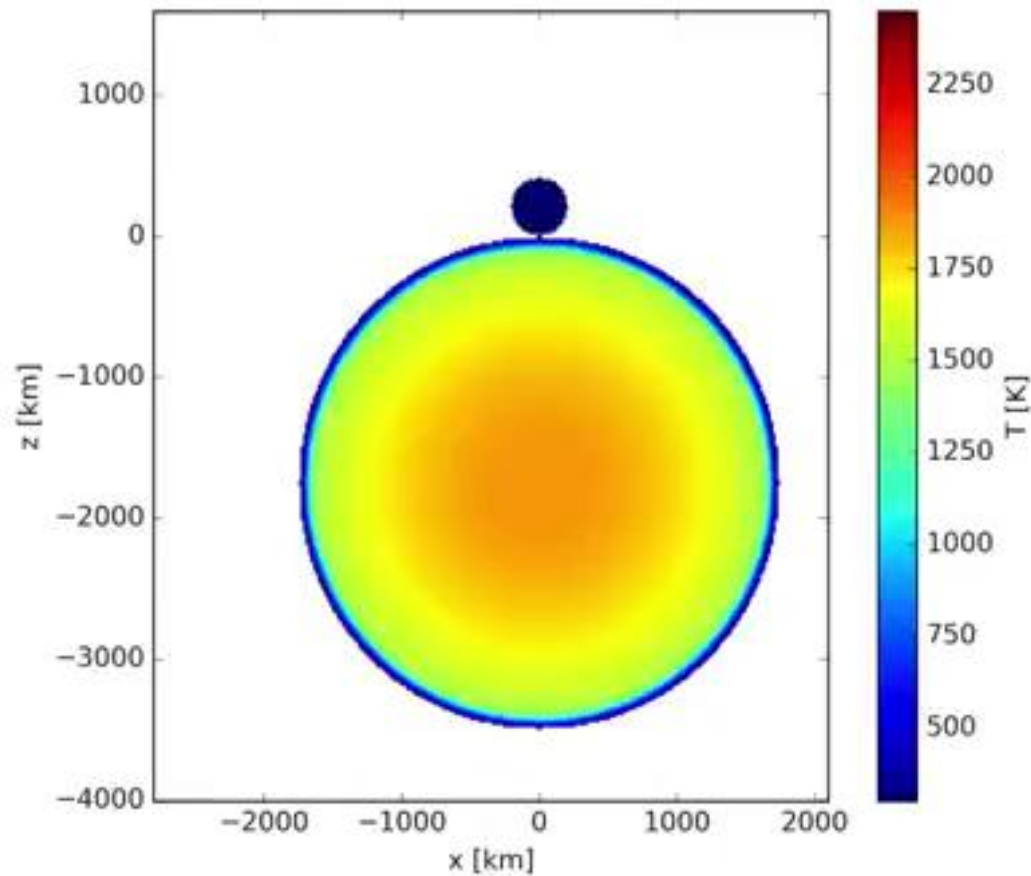
AI-developed models are becoming more prevalent, but may fail to develop formulations based on physical reasoning



Images generated by Perchance using the prompt: AI generated predictive model that has no physical basis and is inaccurate when extrapolated

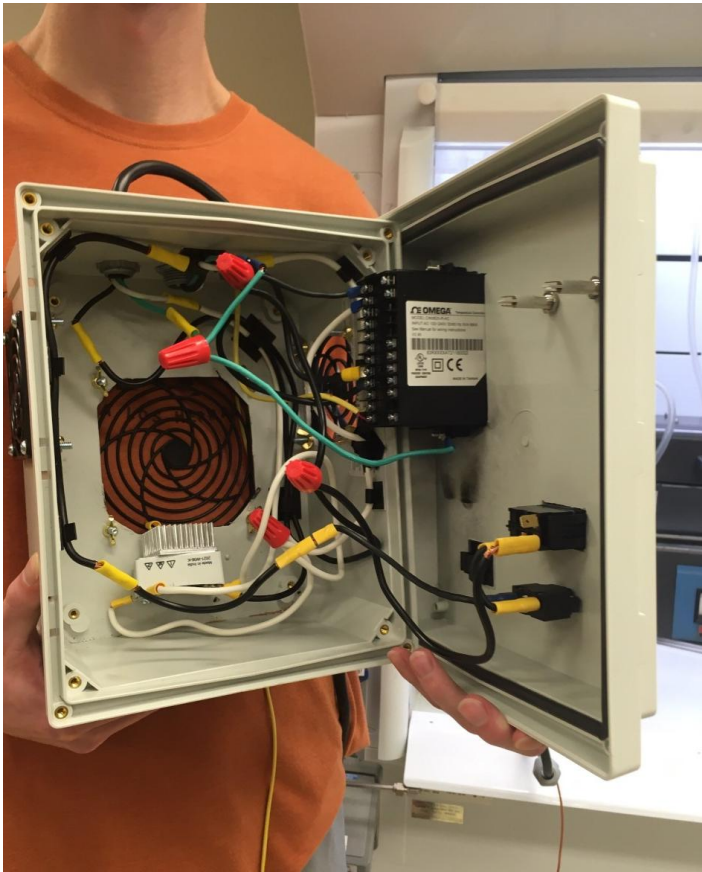
Successfully generalized models can be implemented in dynamic simulations of planetary evolution

Lunar mantle dynamic simulations investigating mantle convection after a giant impact



Only in the context of painstaking experimental work and method development is it possible to make predictions that help us understand the rock record

An experimental furnace control device developed as an undergraduate project at the University of Tennessee (left), and the outcome of a starting material conditioning run (right)



Experiments produce results that faithfully record their conditions

Quenched glass produced in an experiment at the University of Tennessee



All models (including experimentally parameterized models) are wrong, but some are useful

